Comparing U.S. News Topic Trends on Yahoo News, Reuters, and Google Trends

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1. Motivation and Rationale

My project is about comparing U.S. news topic trends on Yahoo News and Reuters, and then examining whether those highlighted topics also align with broader public interest on Google Trends. I chose this topic because I want to understand not only what kinds of news topics are emphasized by different news platforms, but also whether those topics actually reflect public interest. Therefore, my main research questions are:

First, what types of topics appear most frequently on the U.S. pages of Yahoo News and Reuters?

Second, how did topic coverage differ between Yahoo News and Reuters?

Third, do the topics highlighted by these two platforms also align with broader public interest as reflected in Google Trends?

Fourth, which topics show the largest gap between media coverage and public interest?

2. Description of Data Sources

a. Specify exact data sources

The first data source was Yahoo News U.S. I collected article-level data from Yahoo's U.S. news pages (https://news.yahoo.com/us/), including article titles, summaries, publication times, and article URLs. Since Yahoo News did not provide a simple public API for this project, I used Python requests and BeautifulSoup to scrape and parse the HTML pages. After cleaning and filtering the data, the final Yahoo dataset contained 561 articles from Apr 3 to Apr 10, 2026.

The second data source was Reuters U.S. news. I collected similar article-level data from Reuters' U.S.news pages (https://www.reuters.com/world/us/), including article titles, summaries, publication times, and article URLs. Because Reuters uses dynamic loading on its U.S. News page, I saved the relevant Reuters HTML content locally and then used BeautifulSoup to parse the local HTML file. After filtering by date, the final Reuters dataset contained 135 articles from April 3 to April 10, 2026.

The third data source was Google Trends. I manually searched comparable topic terms in Google Trends based on the topic categories used in my news classification and downloaded the trend data as CSV files. Then I used Python to read the CSV files and calculate the average trend score for each topic. These scores were used as an indicator of public search interest.

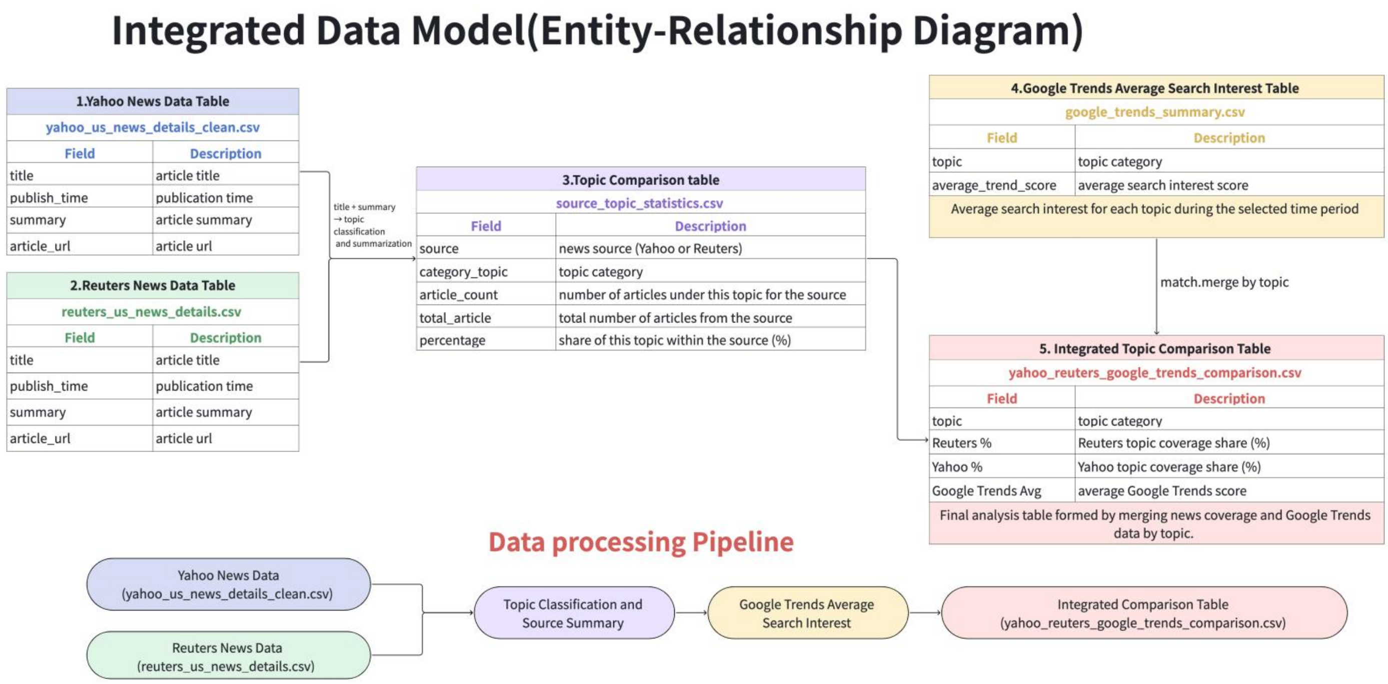
b. Challenges and Changes from the Original Plan

The main challenge was collecting data from Reuters. At first, I planned to use Python requests and BeautifulSoup to directly scrape the Reuters U.S. News page. However, I found that Reuters is a dynamic webpage. Its article list depends on dynamic loading, so a normal requests call could not capture all articles, especially older articles loaded through the "Load More Articles" function.

I also tried to inspect the webpage and use the API request behind the dynamic loading feature, but Reuters restricted direct API access. Therefore, this method did not work for my project.

In the end, I solved the problem by manually loading more articles on the Reuters page, copying the relevant HTML from the browser, saving it locally, and then parsing the saved HTML file with BeautifulSoup. Although this method was not a fully automated crawler, it made the data collection process more stable and reproducible because the original HTML file was saved locally. This also allowed me to keep a fixed dataset for analysis, even if the Reuters webpage changed later.

3. Integrated Data Model



The integrated data model in this project connects article-level news data from Yahoo News and Reuters with topic-level public search interest data from Google Trends. The Yahoo and Reuters datasets both contain the same core fields: title, publish_time, summary, and article_url. After cleaning the two datasets, I combined them into a common news article structure and added a source field to identify whether each article came from Yahoo or Reuters. Since the two datasets had the same fields after cleaning, I combined them using Pandas concat rather than a traditional join.

The title and summary fields were used for topic classification. Each article was assigned to a category_topic, such as Politics, Business, Crime, Law, Science, Health, World News, Environment, Immigration, Education, or Other. After classification, I grouped the articles by source and category_topic to create the Topic Comparison Table. This table contains article_count, total_articles, and percentage, which shows how much each source covered each topic. This produced the source_topic_statistics.csv table.

Google Trends data was collected separately at the topic level. The Google Trends summary table contains topic and average_trend_score, which represents the average public search interest for each topic during the selected time period. Finally, I merged the Topic Comparison Table with the Google Trends summary table by the shared topic field. The final integrated table contains topic, Reuters %, Yahoo %, and Google Trends Avg. This final table allows me to compare media coverage from Yahoo and Reuters with public search interest from Google Trends.

4. Analyses/Visualization

a. **Describe what analysis techniques you used.**

My analysis followed four main steps: cleaning and standardizing the news data, classifying articles into common topic categories, calculating topic percentages for each source, and comparing the results with Google Trends public search interest. After cleaning, the Yahoo dataset saved as yahoo_us_news_details_clean.csv, and the Reuters dataset was saved as reuters_us_news_details.csv. These cleaned files were used as the input for topic classification and later analysis.

First, I used Pandas to clean and standardize the Yahoo News and Reuters datasets so that both sources had the same fields: title, publish_time, summary, and article_url. During the cleaning process, I removed incomplete records, standardized column names, and converted different date formats into a consistent YYYY-MM-DD format. I then filtered both datasets to the selected time period from April 3 to April 10,2026. This step ensured that the yahoo and reuters articles were collected from the same date range, making the later comparison more consistent and fair.

Next, I applied NMF topic modeling as an exploratory step to identify latent themes in the combined Yahoo News and Reuters dataset. This helped me understand the major topic patterns that appeared in the collected articles. However, some NMF topics were too event-specific or overlapping, such as several crime-related subtopics. Therefore, I used the NMF results as a reference and created a shared keyword-based topic classification framework for the final analysis. This framework mapped all articles into a common set of broad topic categories, including Politics, Business, Crime, Law, Science, Health, World News, Environment, Immigration, Education, and Other. For each article, I combined the title and summary into one text field and assigned the article to the category with the highest keyword match score. The final cross-source comparison between Yahoo News and Reuters was based on these common topic labels. After classification, I used Pandas groupby operation to group the articles by source and category_topic to calculate the number of articles in each topic for Yahoo and Reuters. Since Yahoo and Reuters had different total numbers of articles, I calculated topic percentages instead of only comparing raw article counts. The percentage was calculated as article_count divided by total_articles for each source.

For Google Trends, I used pandas to filter the trend data to the same time period and calculated the average trend score for each topic. This average score was used as an indicator of relative public search interest.

Finally, I used a Pandas merge operation to combine the news topic percentage table with the Google Trends summary table by topic. This allowed me to compare media coverage from Yahoo and Reuters with public search interest from Google Trends.

b. Describe the figures

I used Pandas to prepare the summary tables and Matplotlib to create the visualizations.

Figure1：Yahoo vs Reuters Topic Percentage Comparison

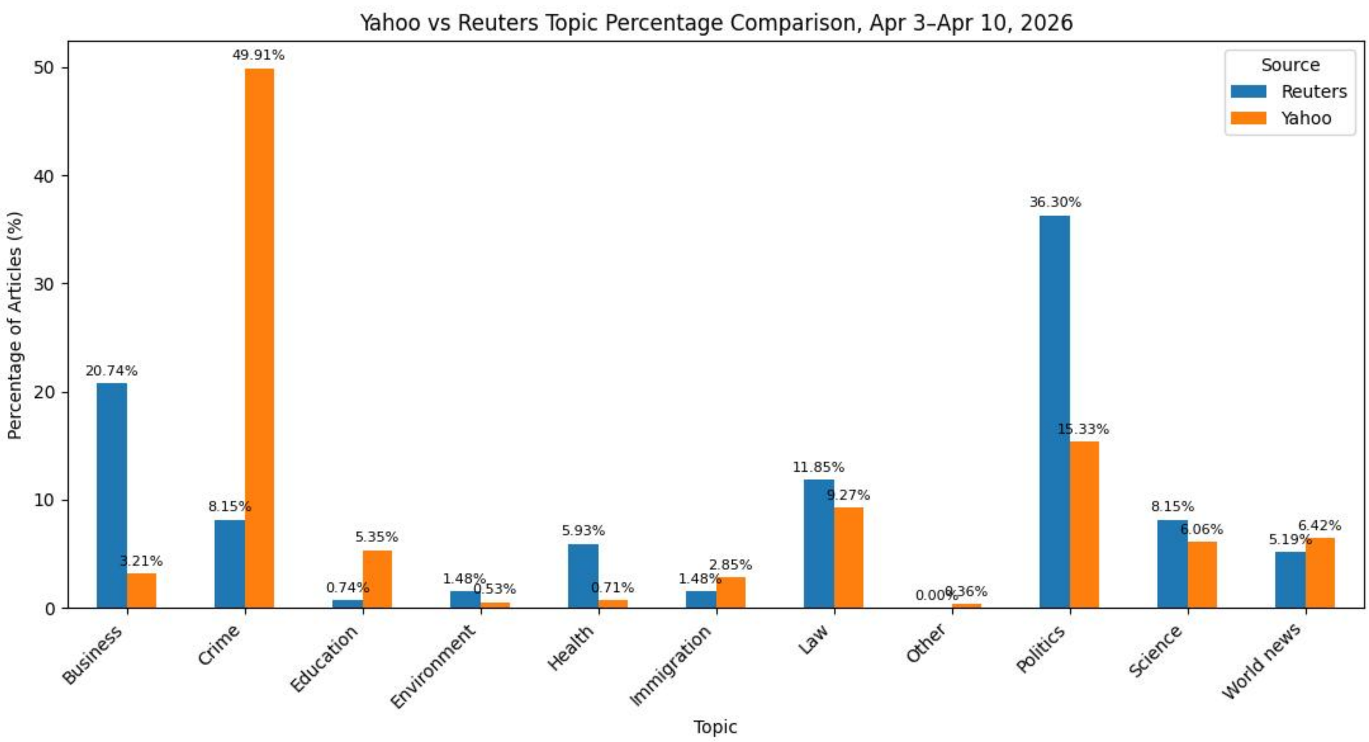


Figure 1 shows the percentage of articles in each topic category for Yahoo News and Reuters. The x-axis represents the topic categories, and the y-axis represents the percentage of articles within each source. Each topic has two bars, one for Yahoo and one for Reuters. I used percentages instead of raw article counts because Yahoo and Reuters had different total numbers of articles. This figure shows that Reuters had a higher share of Politics and Business articles, while Yahoo had a much higher share of Crime and Politics articles.

Figure 2. Average Google Trends Public Interest by Topic

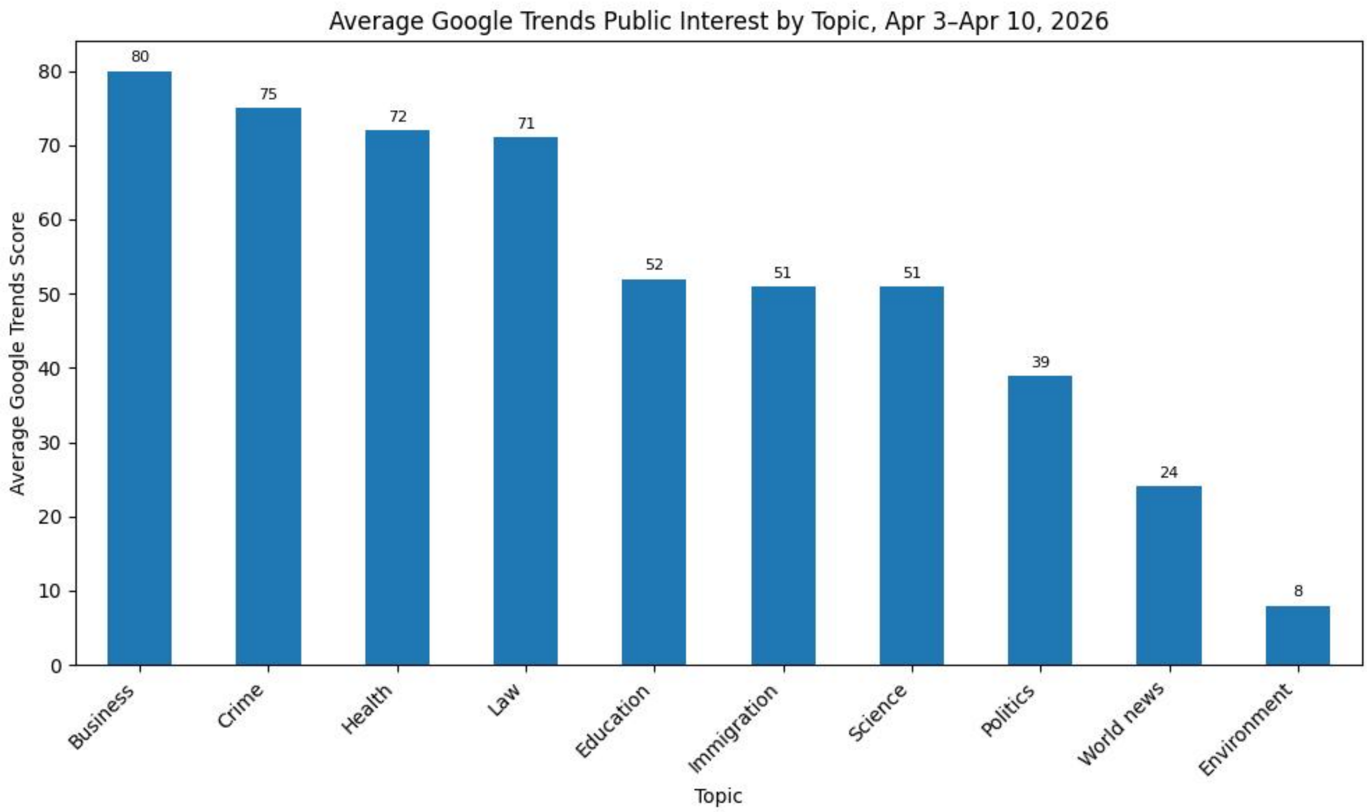


Figure 2 shows the average Google Trends score for each topic from April 3 to April 10, 2026. The x-axis represents the topic categories, and the y-axis represents the average Google Trends score. This score is a relative search interest index, not a raw search count. The figure shows that Business, Crime, Health, and Law had the highest average public search interest during the selected period.

Figure 3. News Coverage and Google Trends Comparison

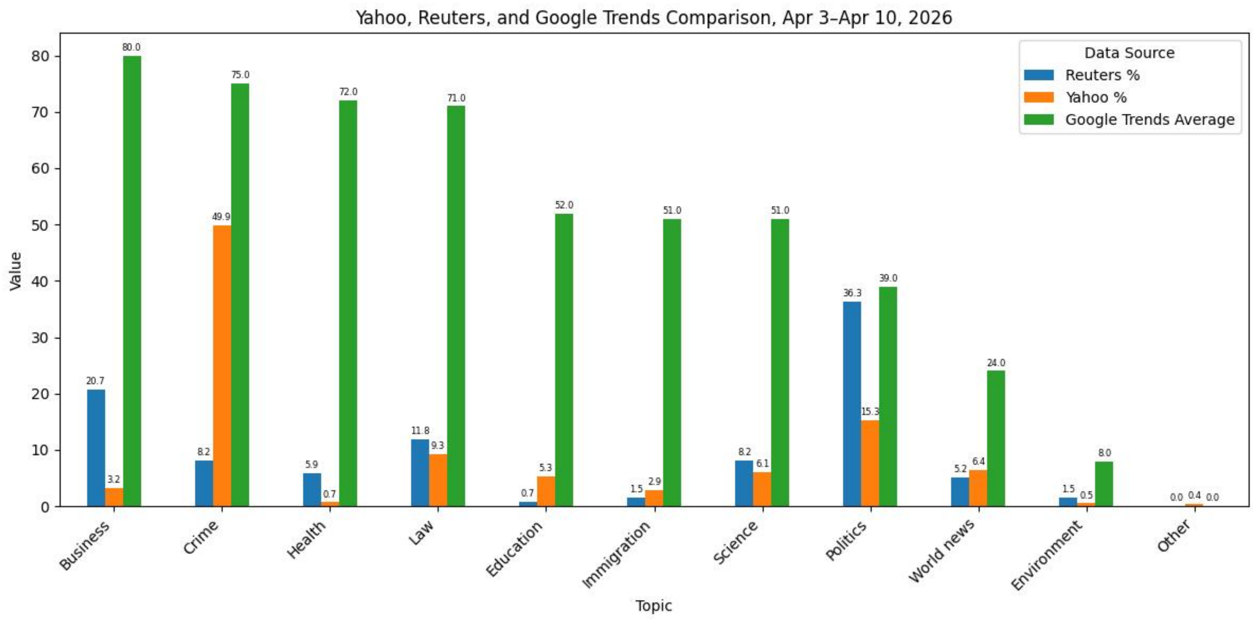


Figure 3 combines Reuters topic percentage, Yahoo topic percentage, and Google Trends average score in one grouped bar chart. The x-axis represents topic categories. The y-axis represents values on a 0 – 100 scale, but the three bars have different meanings: Reuters and Yahoo bars show news coverage percentages, while the Google Trends bar shows the average relative search interest score. This figure helps compare whether media coverage patterns were aligned with public search interest. （Google Trends Average is not a percentage; it is a relative search interest index.）

5. Conclusions

Based on the analysis above, this project answers the four research questions as follows.
First, Politics was the most important shared topic across both platforms because it appeared frequently in both Yahoo News and Reuters. However, the two platforms still had different main priorities. Reuters focused most on Politics, while Yahoo News focused most on Crime.

Second, topic coverage differed clearly between Yahoo News and Reuters. Reuters had a higher percentage of Politics (36.30%) and Business (20.74%) articles, while Yahoo had a much higher percentage of Crime (49.91%) articles. This suggests that Reuters focused more on institutional and policy-related news, while Yahoo placed more emphasis on incident-based stories, including crime, accidents, and public safety events.

Third, the relationship between news coverage and public search interest was only partly consistent. Google Trends showed the highest average search interest in Business, Crime, Health, and Law during the selected period. Yahoo’s strong focus on Crime was relatively aligned with Google Trends, because Crime also had high public search interest. Reuters’ coverage of Business was also partly aligned with Google Trends, since Business had the highest average trend score. However, Reuters focus on Politics, while Politics did not have the highest Google Trends score. This suggests that media coverage priorities and public search interest were not completely consistent.

Fourth, Business and Politics showed the clearest gaps between media coverage and public interest. Business ranked highest in Google Trends, but it received limited coverage on Yahoo. Politics was the dominant topic in Reuters, but it did not rank highest in Google Trends. These results suggest that news coverage did not always match public search interest.

In conclusion, this project found that Yahoo News and Reuters covered U.S. news topics differently during April 3 – April 10, 2026. Reuters emphasized Politics and Business, while Yahoo emphasized Crime. Google Trends suggested that public search interest was strongest for Business, Crime, Health, and Law. Therefore, news coverage and public interest were only partially aligned, and the degree of alignment varied by topic and by news platform.

6. Future work

Given more time, I would improve this project in several directions. First, I would expand the data collection period. This project only focused on articles from April 3 to April 10, 2026, so the results may reflect short-term news events. A longer time period would make the topic comparison more stable and representative.

Second, I would improve the topic classification method. In this project, I used NMF topic modeling as an exploratory step and then used a keyword-based framework for the final classification. In the future, I would test more advanced NLP methods, such as supervised text classification or BERTopic, and manually validate a sample of classified articles to measure classification accuracy.

Finally, I would include more news platforms, such as CNN, The New York Times. This would make the comparison broader and help determine whether the hot topics found in Yahoo News and Reuters were unique to these two platforms or were also commonly covered by the wider media.